

EXPERIMENT – 30

Implementation of CBC-MAC Forgery for a Two-Block Message

CODE:

```
#include <stdio.h>

#define N 8

// Simple block cipher for demonstration
void encrypt(int in[], int out[], int key[])
{
    for (int i = 0; i < N; i++)
        out[i] = in[i] ^ key[i];
}

void xorBlock(int a[], int b[], int result[])
{
    for (int i = 0; i < N; i++)
        result[i] = a[i] ^ b[i];
}

void printBlock(int a[])
{
    for (int i = 0; i < N; i++)
        printf("%d", a[i]);
}

int main()
```

```

{
    int X[N] = {1,0,1,0,1,0,1,1};
    int K[N] = {1,1,0,0,1,1,0,0};

    int T[N], block2[N], temp[N], mac2[N];

    /* One-block CBC-MAC */
    encrypt(X, T, K);

    /* Second block = X XOR T */
    xorBlock(X, T, block2);

    /* CBC-MAC for X || (X XOR T) */
    xorBlock(T, block2, temp);
    encrypt(temp, mac2, K);

    printf("Message X      = ");
    printBlock(X);

    printf("\nMAC(K,X) T      = ");
    printBlock(T);

    printf("\nSecond Block    = X XOR T = ");
    printBlock(block2);

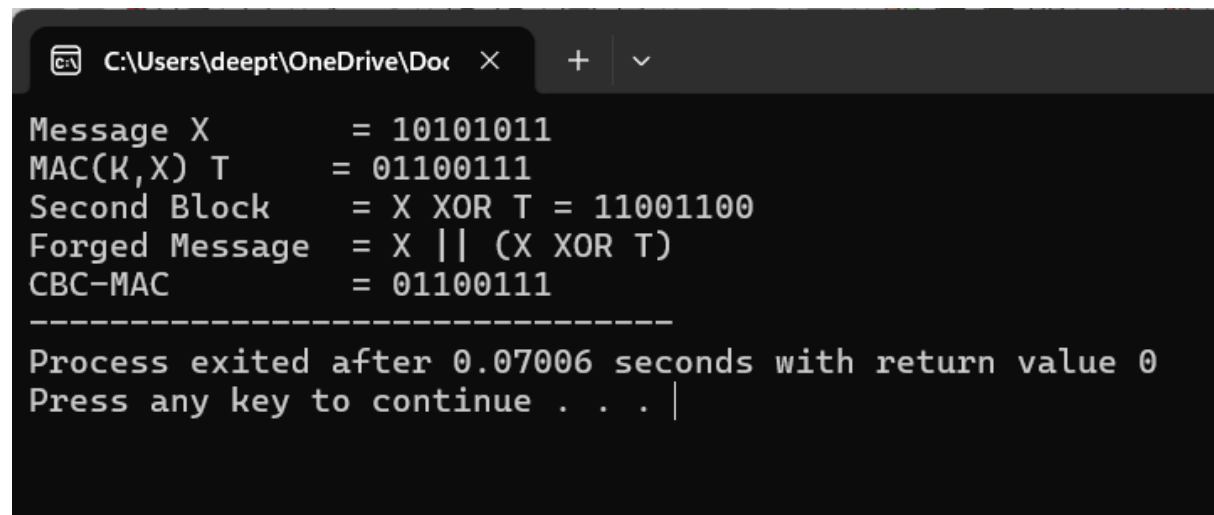
    printf("\nForged Message = X || (X XOR T)");

    printf("\nCBC-MAC          = ");
    printBlock(mac2);

```

```
    return 0;
}
```

OUTPUT:



```
C:\Users\deept\OneDrive\Doc >
Message X      = 10101011
MAC(K,X) T     = 01100111
Second Block   = X XOR T = 11001100
Forged Message = X || (X XOR T)
CBC-MAC        = 01100111
-----
Process exited after 0.07006 seconds with return value 0
Press any key to continue . . . |
```